

Homework for Module on Dimensionality Reduction

1. Fisher's linear discriminant is

$$\hat{\mathbf{w}}^* = \arg \max_{\hat{\mathbf{w}}} J(\hat{\mathbf{w}}) = \arg \max_{\hat{\mathbf{w}}} \frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}},$$

where $\mathbf{S}_b = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$ and $\mathbf{S}_w = \sum_j \sum_{\alpha} (\mathbf{x}_{\alpha} - \mathbf{m}_j)(\mathbf{x}_{\alpha} - \mathbf{m}_j)^T$.

- (a) By writing $\frac{dJ(\hat{\mathbf{w}})}{d\hat{\mathbf{w}}} = 0$, show that

$$\mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} = J(\hat{\mathbf{w}}) \hat{\mathbf{w}}.$$

- (b) Explain why $\hat{\mathbf{w}}^*$ is the eigenvector for which $J(\hat{\mathbf{w}})$ is the maximum eigenvalue of $\mathbf{S}_w^{-1} \mathbf{S}_b$.
 (c) Explain why $\mathbf{S}_b \hat{\mathbf{w}}$ is always in the direction of $\mathbf{m}_1 - \mathbf{m}_2$ and thus show that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1} (\mathbf{m}_1 - \mathbf{m}_2).$$

2. Another way to optimize Fisher's linear discriminant (suggested by Barry Fridling):

- (a) Show that for any two real vectors $\mathbf{x}$ and $\mathbf{y}$

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{x})(\mathbf{y}^T \mathbf{y}), \quad (\text{Cauchy-Schwarz}).$$

- (b) Show that if the λ_k are positive,

$$\left(\sum_{k=1}^N x_k y_k \right)^2 \leq \left(\sum_{k=1}^N \lambda_k x_k^2 \right) \left(\sum_{k=1}^N y_k^2 / \lambda_k \right).$$

- (c) Thus, show that for $\mathbf{A}$ positive definite

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{A} \mathbf{x})(\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y}).$$

- (d) By letting $\mathbf{A} = \mathbf{S}_w$ in the expression above, and writing

$$J(\hat{\mathbf{w}}) = \frac{|\hat{\mathbf{w}}^T (\mathbf{m}_1 - \mathbf{m}_2)|^2}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}},$$

show again that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1} (\mathbf{m}_1 - \mathbf{m}_2).$$

3. Using the Optdigits dataset from the UCI repository, implement PCA. Reconstruct the digit images and calculate the error $E(n) = \sum_{j=1}^m \|\hat{\mathbf{x}}_{j,n} - \mathbf{x}_j\|^2$ for various values of n , the number of principal components included in the reconstruction of each image. Plot $E(n)$ versus n .